Research on Hybrid Online Brain-Computer Interface System Based on MI-mVEP

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***Abstract*—In this study, a hybrid online brain-computer interface (BCI) system was established to address the issue of low recognition accuracy and limited imagery tasks in motor imagery (MI) based on BCI systems. This hybrid system utilized both MI and motion-onset visual evoked potential (mVEP). After simultaneously acquiring MI and mVEP potentials and electroencephalogram (EEG) signal preprocessing, the filter bank common spatial pattern (FBCSP) algorithm was used to extract MI and mVEP features. These features were then employed to construct training set feature vectors. The training set feature vectors of MI and mVEP signals were classified using the backpropagation (BP) neural network and support vector machine (SVM) algorithms, respectively. Based on the established classification model, single test data were recognized online, and the results were fed back to subjects in real-time via block moving routes. The presented system can realize five different block moving paths, with faster moving effects achieved when the system induces MI and mVEP potential simultaneously along the lower left or lower right diagonal path. The experimental results showed that the average classification accuracies of the lower left and lower right diagonal moving effects are up to 84.53% and 84.60%, respectively. These results indicate that this study effectively controls the fast-moving blocks in the online BCI system of MI-mVEP, providing a theoretical basis and experimental support for the research of a hybrid online BCI system based on MI-mVEP. In summary, the proposed system has demonstrated improved performance through the combination of simultaneous MI and mVEP potential utilization and efficient classification algorithms, with promising future applications in BCI research.**

Keywords-MI; mVEP; Brain-computer interface; Online control; Hybrid system

I. INTRODUCTION

The brain-computer interface (BCI) is a new way of communication and control between the human brain and computers or other devices [1]. As a typical BCI paradigm, motor imagery (MI) involves motor intention but no actual output, resulting in event-related desynchronization (ERD) and event-related synchronization (ERS) of the mu and beta rhythms of the sensorimotor cortex. MI-BCI is an active BCI paradigm that does not require external stimulation and reflects the motor awareness of users. It has played a crucial role in compensating for motor dysfunction and inducing brain plasticity [2].

Motion-onset visual evoked potential (mVEP) is the electric activity of the cerebral cortex that is produced in response to human vision stimulated by the rapid onset of external movement [3]. The visual evoked potential occurs mainly in the middle temporal area and medial superior temporal area of the brain [4]. mVEP has been gaining attention in the BCI field because the evoked potential itself is generated according to movement without the need for light changes, frequency flickers, or other strong visual stimuli forms, which reduces subject fatigue during the experiment [5-8]. Additionally, since the human eyes are more sensitive to moving objects, the mVEP experiments can obtain higher stability. The mVEP has three characteristic amplitude waveforms, P1, N2, and P2. Among these waveforms, the N2 component has been identified as the main feature of the mVEP signal. The N2 component refers to a waveform with a descending crest that primarily appears within 160-200ms after the start of visual stimulation, which is a significant signal feature of the mVEP potential [9-10]. Currently, there are four common evoked modes in mVEP research, which are circle diffusion movement evoked, circle contraction movement

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evoked, horizontal movement evoked, as well as up and down movement evoked. Compared with steady-state visual evoked potential (SSVEP) and steady-state somatosensory evoked potential (SSSEP), the mVEP has more apparent amplitude changes, which can be analyzed quickly in the feature recognition process [11].

In recent years, there have been growing reports on mVEP BCI, especially the combination of mVEP with other BCI paradigms. In 2019, Beveridge Ryan et al. combined mVEP signals with games and developed a first-person shooter game application software using mVEP-BCI, achieving recognition accuracy above 90%, which ensured the fluency of game operation [12]. In 2020, Ma et al. integrated MI and mVEP to create a hybrid BCI system. The average classification accuracy of the system was about 76%, and the transmission rate was 13 bits/min, which enabled a more natural and effective control of the cursor in a more comfortable mode [13]. However, this study only carried out a simple fusion of MI and mVEP and did not explore a more effective mixing mode of the two paradigms. Moreover, the classification accuracy of the online system was not satisfactory. Therefore, this study proposes a more reasonable and natural hybrid paradigm to design an MI-mVEP online system to improve the classification accuracy of the system. Since mVEP signals are highly inducible and the inducement mode is relatively safe to avoid the risk of disease, this study attempts to combine MI and mVEP into a hybrid online system to increase the number of controllable tasks and improve the control efficiency of the online system with a more natural interactive way of BCI system.

II. SYSTEM BUILDING AND EXPERIMENT DESIGN

The MI-mVEP hybrid paradigm of the online system includes both software and hardware components. The hardware components of the system consist of the NeuroScan 64-channel EEG acquisition amplifier, the international standard 10-20 system, and the computer that runs the experimental stimulus paradigm and performs online data processing. The software part of the system includes the experimental stimulus paradigm and experimental feature extraction and classification algorithm written based on MATLAB PTB toolbox. The MI-mVEP online system designed in this study mainly consists of five modules: EEG acquisition, preprocessing, feature extraction, classification, and animation feedback. The block diagram of the MI-mVEP hybrid online system is shown in Figure1.

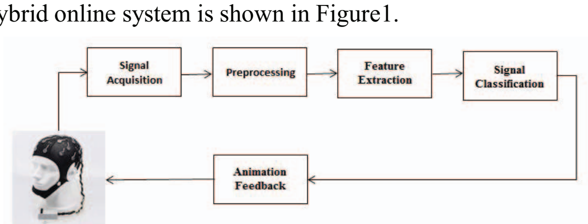


Figure 1. Flow chart of MI-mVEP hybrid online system

To design the MI-mVEP online experiment paradigm, the experiment was divided into two parts: the task training experiment for constructing the MI-mVEP classification model and the real-time test experiment for the MI-mVEP task. The training task mode is illustrated in Figure 2, where each trial lasts for 6 seconds, with the first second dedicated to preparation. During the 1-5 second hybrid task time, subjects need to concentrate on receiving the normal stimulus (MI, mVEP, or both). The task ends during the 5-6 second period, and subjects can take proper rest to reduce fatigue during the experiment. In the MI-mVEP task training stage, subjects were required to perform three groups (42 trials) of MI-mVEP tasks, where the mVEP stimulation only needed to be fixed, while the left and right hand motor imagery task needed to be completed according to the task prompts. In each group of tasks, the left and right hand movements were performed 21 trials equally. During the experiments, the experimental data were collected using a 64-lead EEG cap, recorded and saved by SynAmps2 amplifier and Scan4.5 software, and transmitted to another PC for data modeling and classification through the TCP/IP protocol. The signals of MI and mVEP hybrid paradigm were processed separately to ensure accurate analysis and model construction.

Figure 2. Schematic diagram of the MI-mVEP training mission mode

For the MI-mVEP test task, the mode diagram is shown in Figure 3. Subjects complete the corresponding operation according to the screen prompts in the experiment, and the screen will give real-time feedback display according to the performance of the task.

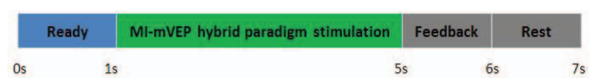
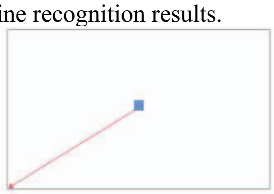
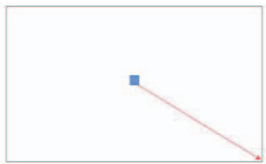


Figure 3. Schematic diagram of MI-mVEP test task

The feedback function mainly presents the online recognition results to the subjects in an intuitive and visual form. In this study, the recognition results of MI and mVEP were presented in the form of numbers, where 1 represented the left-oriented MI task, 2 represented the right-oriented MI task, and 3 represented the mVEP task. The MI-mVEP online system uses the form of pictures to give feedback to make the subjects more intuitively accept the feedback information. The feedback time of pictures is 1s.When the recognition result of the MI-mVEP BCI system is only "1", the feedback terminal of PC will represent the picture of "the block moving to the left," as shown in Figure 4(c). When the recognition result of MI-mVEP is only "2", the picture of "block moving to the right" appears in the feedback terminal of PC, as shown in Figure 4(d). When the result of the MI-mVEP system is only "3", the feedback terminal will show the picture of "block moving down," as shown in Figure 4(e). The above three situations are the feedback effect when the online system recognizes only one type of EEG. When the MI-mVEP online system recognizes both types of MI and mVEP, the feedback terminal will display a picture of "square moving along the lower left or right diagonal" according to the current recognition results, as shown in Figure 4(a) or 4(b). For example, when the recognition results are "2" and "3", the image effect of "square moving diagonally along the lower right" will be displayed on the PC terminal, as shown in Figure 4(b). In this online experiment, threshold values are set for the three recognition results of "1", "2", and "3". Only when the online data analysis results reach the threshold, the system can output corresponding values. For the purpose of the block moving along the diagonal, the system can reduce the number of moving steps under the fixed distanced to shorten the movement time. Figure 4 shows the feedback diagram of online recognition results.

(a) (b)


(c) (d)

(e)

Figure 4. The effect of online recognition result feedback

In this study, a total of 10 participants were recruited to collect EEG signals through an experiment. The group consisted of 4 females and 6 males, with an average age of 24.2 ± 1.2 years, and all participants were in good health. The first three groups underwent training tasks during which they performed a total of 3*42 trials, while configuring the classification model and spatial filter. The time taken to construct the model ranged between 0.16s and 0.20s. During the testing task, the MI-mVEP task was completed via 42 trial tasks. Real-time display of block movements on the PC monitor was employed in this study as feedback to participants regarding the effectiveness of their trial execution. This feature was a key component of the MI-mVEP online system, which facilitated the real-time adjustment of experimental states based on performance feedback.

III. ONLINE REAL-TIME DATA PROCESSING OF UPPER LIMB MI-MVEP EEG SIGNALS

A. *Preprocessing of EEG signal*

The EEG data preprocessing in this study was conducted using the EEGLAB toolbox. Firstly, electrodes unrelated to MI tasks were removed to improve the overall processing speed of the system. Then, the 60 leads related to the cerebral cortex were located using the 10-20 electrode standard system to analyze the spatial feature information in the EEG data of the upper limb. The collected EEG data was re-referenced to ensure the accuracy of data processing. Band pass filtering was used to reduce noise in the EEG data, and 1Hz cut-off High-pass filtering and 75Hz cut-off Low-pass filtering were needed for MI data to remove noise interference. The data was segmented according to the task period set in the experimental paradigm, and the data of the rest period was excluded. Baseline correction and removal of bad segments from EEG data were completed using the EEGLAB toolbox. Finally, independent component analysis (ICA) was run to conduct unified artifact processing on the collected EEG data of upper limbs, including the removal of irrelevant signals such as electromyography or eye movement and blink artifacts.

B. *Feature extraction algorithm*

The FBCSP algorithm was utilized in this study to extract features from MI and mVEP signals. This optimization algorithm is based on the CSP algorithm and processes information in various frequency bands. FBCSP has a more effective feature extraction effect when MI signals are dispersed throughout different frequency bands, such as the alpha band (8-13Hz) and beta band (14-30Hz). In the FBCSP algorithm, the third-order Butterworth filter was employed to analyze the MI EEG signals based on their respective feature frequency bands of 0.5~10Hz, 10~14Hz, 11~17Hz, 14~20Hz, 17~23Hz, and 20~26Hz. The first 5 seconds of data from each trial were filtered according to these 5 frequency bands, and signals in each band were extracted by utilizing FBCSP to calculate the mutual information of FBCSP features in each band. The MI and mVEP signals were separated based on their frequency difference, and the features with the highest mutual information were independently input into a BP neural network and SVM classifier for online recognition of MI and mVEP.

C. *Classification algorithm*

1) *BP neural network*

In the MI-mVEP online BCI system, the FBCSP+BP neural network algorithm is used to enhance the accuracy of recognition and ensure the stable execution of MI tasks. The BP neural network consists of three layers: input layer, hidden layer, and output layer. The input layer contains 420 neurons, based on the number of feature values obtained from MI after feature extraction by FBCSP. The hidden layer's number of neurons is determined by adjusting the BP neural network. The output layer has only one neuron and outputs the results of the "left-handed" and "right-handed" tasks in MI classification. Figure 5 indicates the structure diagram of a three-layer neural network.

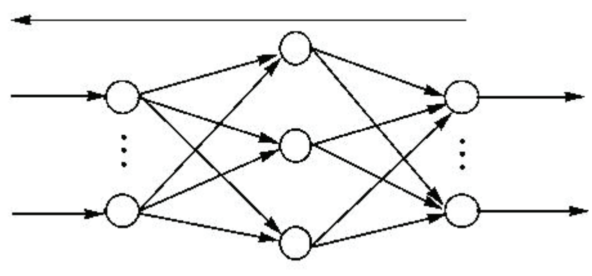


Figure 5. Structure diagram of a three-layer neural network

2) *SVM algorithm*

In this study, the FBCSP+SVM algorithm was employed to ensure the stability and high accuracy of the MI-mVEP system as mVEP signals contain less information to be identified. The

SVM model has a relatively short predictive time, making processing of mVEP signals on PC more stable and reducing the overall feedback time of the system. Nonlinear SVM maps the sample space to a high-dimensional or infinite-dimensional feature space through nonlinear mapping, which solves the problem of nonlinear separability in the original sample space. The algorithm flow is as follows:

a) Input EEG training set data, as shown in (1) :

$$T = \{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)\} \quad (1)$$

Where is EEG data, represents categories of different EEG tasks, .

b) Select kernel function $K(x, y)$ and appropriate parameter C to construct the corresponding function as shown in (2) :

$$\min_{\alpha} \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j K(X_i X_j) - \sum_{i=1}^{N} \alpha_i$$

$$s.t \sum_{i=1}^{N} \alpha_i y_i = 0 \quad (2)$$

$$0 \leq \alpha_i \leq C, i = 1,2, \cdots, N$$

The optimal solution is obtained, as shown in (3) :

$$\alpha^* = (\alpha_1^*, \alpha_2^*, \cdots, \alpha_n^*)^T \quad (3)$$

c) Select α^* *integral component* $0 < \alpha_n < C$ *,parallel calculation* b^**, as shown in (4) :*

$$b^* = y_i - \sum_{i=1}^{N} \alpha_i^* y_i K(x_i, y_j) \quad (4)$$

d) Construct the decision function $f(x)$*, as shown in (5) :*

$$f(x) = sign(\sum_{i=1}^{N} \alpha_i^* y_i K(x_i, y_j) + b^*) \quad (5)$$

When $K(x, y)$ is a positive definite function, it becomes a convex quadratic programming problem, and the optimal solution is obtained.

During the early training task, the features of the MI and mVEP training sets that were extracted by FBCSP were input separately into the BP neural network and SVM classifier to execute the modeling of the training sets. In the subsequent test experiment, the EEG data of each trial was transmitted to the computer for online data processing via the TCP/IP protocol. The extracted features of the MI and mVEP signals after FBCSP were input into their respective training set models for prediction, and the prediction results were provided to the subjects in real-time via feedback.

D. Characteristic analysis method of EEG signal

a) Time-frequency analysisSelection

The time-frequency analysis can present the changes in time and frequency of upper limb MI data in the form of images to clearly observe the data changes. The upper limb MI signal studied in the experiment has the characteristic of energy attenuation in a specific time and frequency band. The short-time Fourier transform method is used to calculate the characteristic frequency band changes in the selected time period. The formula for the short-time Fourier transform is shown in (6), where t and f represent the time and analysis frequency, respectively, where $x(u)$ is the data information for a single experiment to be calculated, and $g(u - t)$ is the fixed observation window function.

$$STFT(t, f) = \int_{-\infty}^{\infty} x(u) * g(u - t) e^{-j2\pi f u} du \quad (6)$$

The disturbance value of the event-related spectrum is represented by the average energy value of short-time Fourier transform results and its formula is shown in (7):

$$ERSP(t, f) = \frac{1}{m}\sum_{p=1}^{m} (STFT_k^2(t, f)) \quad (7)$$

Where: m represents the number of subjects collected for the EEG experiment, and $STFT_k^2(t, f)$ represents the estimated frequency spectrum of the trials at the selected time and frequency.

b) Brain topographic map analysis

The brain topographic map is a comprehensive image of the brain constructed by utilizing the ERSP value. With the lead number and ERSP value, a spatial distribution map of the ERD feature is formed. The diversity of ERSP values across various electrodes gives rise to distinct brain topographic maps. For the MI data analysis carried out in this study, the ERD characteristics of the C3 and C4 electrodes were primarily examined, with focus on the alpha and beta bands as selected feature bands.

IV. MI-mVEP EXPERIMENTAL RESULTS AND ANALYSIS

A. Accuracy analysis of upper limb MI-mVEP signal recognition

Data processing for the MI-mVEP online brain-computer interface system mainly involves EEG preprocessing and FBCSP feature extraction. The BP neural network and support vector machine are utilized to execute online classification of MI and mVEP data, respectively. Real-time feedback is then provided through an animated visual display of moving blocks.

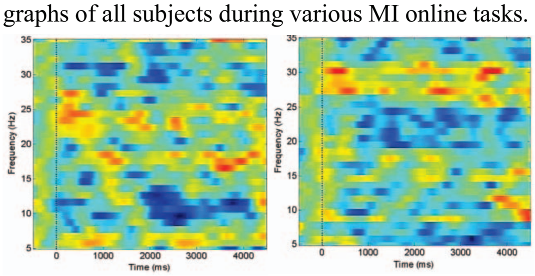
The upper limb MI-mVEP online BCI system offers immediate feedback of the test results to the subjects while also recording the MI task prompt label and feedback outcomes for each trial. In our system, the motion trajectory of both MI and mVEP signals are combined to generate a "diagonal" movement trajectory only when both types of signals are successfully induced. As shown in Table 1, the average classification accuracies of left hand, right hand, and mVEP are all over 80%, with the highest reaching over 90%. In the MI-mVEP hybrid paradigm, the online BCI system's feedback accuracy of "block move along the lower left diagonal" and "block move along the lower right diagonal" were also above 80%, with average accuracies of up to 84.53% and 84.60%, respectively. All subjects in the MI-mVEP online BCI system demonstrated high success rates in triggering the feedback effect of "block moves along the diagonal," and the ITR of all subjects exceeded 15bits/min, ensuring normal operations of the online system.

TABLE I. MI-MVEP ONLINE RECOGNITION ACCURACY (%)

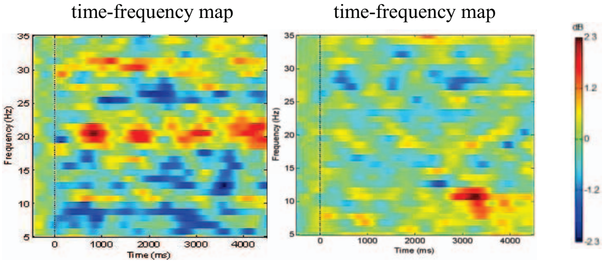
Subject	Left-handed	Right-handed	MI-mVEP	moving along the lower left diagonal	moving along the lower right diagonal	ITR (bits/min)
S1	77.50	82.50	87.50	82.50	80.75	16.68
S2	82.50	80.00	89.25	83.75	84.50	18.56
S3	77.50	75.00	90.00	80.50	81.25	19.23
S4	75.00	80.00	83.75	81.50	83.75	20.16
S5	90.00	92.50	95.50	92.75	90.00	17.38
S6	92.50	85.00	89.50	89.00	87.50	18.55
S7	80.00	87.50	87.00	84.25	85.00	18.42
S8	85.00	80.00	75.00	80.00	81.50	19.77
S9	82.50	90.00	92.25	88.25	87.25	15.84
S10	85.00	77.50	88.75	83.75	84.50	16.69
Mean ±	82.75±3.11	83.00±3.16	87.85±3.01	84.53±1.87	84.60±1.56	18.13 ±1.84

B. Characteristics analysis of MI-mVEP signals in upper limbs

In the MI-mVEP online system, the MI signal features comprise of the ERD features of both left and right MI tasks. In order to validate the results of the MI-mVEP online system, time-frequency analysis and brain topographic map were utilized to conduct feature analysis on both MI and mVEP signals offline. Figure 6 depicts the average time-frequency graphs of all subjects during various MI online tasks.



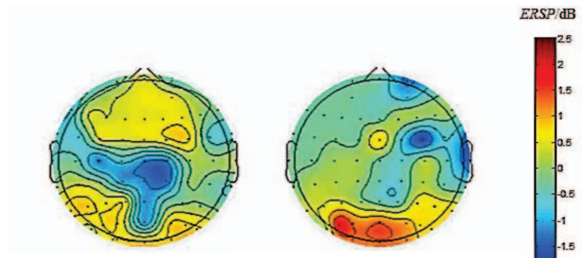
(a)Left-hand task C3 lead time-frequency map (b) Left-handed task C4 lead time-frequency map



(c)Right-hand task C3 lead time-frequency map (d) Right-handed task C4 lead time-frequency map

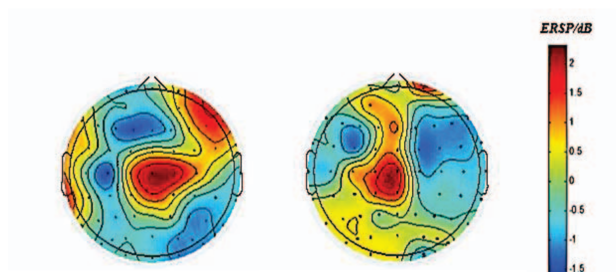
Figure 6. Average time-frequency graphs of all subjects in different MI online tasks

As evidenced by Figure 6, significant ERD phenomena occur in the alpha band (8~13Hz) and beta band (14~30Hz), indicating that the subjects successfully completed the corresponding MI tasks, thus affirming that the online MI-mVEP system can successfully induce MI features. Additionally, in the left and right MI tasks, the time-frequency graph of the left hand MI task exhibits a stronger ERD phenomenon than that of the right hand MI task.



(a) alpha band right hand mission (b) alpha band left hand mission

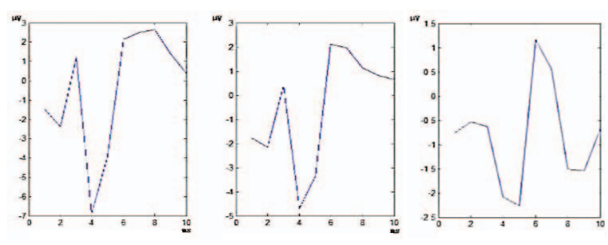
Figure 7. Average brain topographic map of all subjects at C3 and C4 leads in alpha band



(c) beta band right hand mission (d) beta band right hand mission

Figure 8. Average brain topographic map of all subjects at C3 and C4 leads in beta band

Figure 7 and Figure 8 display the average brain topographic maps of ten subjects in the alpha and beta bands during the MI-mVEP task. Based on the average brain topographic map analysis, significant ERD phenomena in the C3 and C4 leads were observed when the subjects performed different MI tasks, which was consistent with the time-frequency results, and the ERD phenomenon in the alpha band was more pronounced than that in the beta band for the same MI task.



(a) PO6 lead characteristics (b) P4 lead characteristics (c) P6 lead characteristics

Figure 9. Average brain topographic map of all subjects at C3 and C4 leads in beta band

Figure 9 illustrates the analysis of mVEP signal features in the online MI-mVEP experiment using down-sampling and overlay average features. The data collected from three leads

(PO6, P4, and P6) selected between the 150 to 300ms demonstrated the same feature as that of a single mVEP task. Specifically, all three leads exhibited a descending crest at 200ms (N200 signal), which is consistent with the characteristics of a single mVEP task.

V. CONCLUSIONU

This study introduces the MI-mVEP hybrid system for online recognition, outlines the MI-mVEP online stimulus paradigm and experimental tasks, and describes in detail the online data feature extraction and classifier modeling methods. The results of the tests were provided to subjects in real-time using a block moving route. The research demonstrated that the average classification accuracy of the feedback effect for "block moving along the lower left diagonal" and "block moving along the lower right diagonal" in the online BCI system of MI-mVEP was 84.53% and 84.60%, respectively. This study successfully achieved the aim of controlling the quick movement of blocks in the MI-mVEP online BCI system and established the groundwork for precise control of the online BCI system. As a result, this study is expected to accelerate the practical implementation of the BCI system.

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